

Split-brain analysis and proposals for DIL-CU

Personal notes

One essential requirement of the NxMooM architecture is that at any time at most one CN is the master CN. Unavailability of CNs is due to a connection loss or a failure of the CNs themselves. Therefore, a mechanism needs to be put in place to minimize probability (or fully prevent if possible) of the split-brain.

State from 17 Mar 2026

- [Proposal](#)
- [Summary](#)
- [Proposed requirements for IPS](#)
- [Follow up discussions with stake holders \(security, safety, HW, IPS\)](#)

Jira epic: [\[ELFOUR-1340\] Split-Brain in case of 2\(4\)x2oo2 - Jira](#)

Jira analysis:

- [\[ELFOUR-1649\] Analyze Plf Solutions for Split-Brain - Part 1 - Jira](#)
- [\[ELFOUR-1712\] Analyze Plf Solutions for Split-Brain - Part 2 - Jira](#)
- [\[ELFOUR-1713\] Analyze Plf Solutions for Split-Brain - Part 3 - Jira](#)

Customer documents:

- [D6 LH Netze](#)
- [D2.2 Teillastenheft RAMS](#)

Current LAB solution with Master Confirm Interface to handle reboots and recovery during the CG startup in 2x2oo2:

- [\[ELFOUR-509\] Reboot of 2x2oo2 is likely to end up in recovery - Jira](#)

Input documents from TAS Plf:

- [TAS Platform Core Programmers Guide](#)
- [SDD - TAS Platform Core SW - 2.x -NMR](#)
- [Safety Concept \(SACO\) - TAS Platform Core - 2.x -CS/FT/NMR](#)

Generic server cabinet:

- [3BU 16010 AZAA JDAPA](#)

Preliminary availability estimations by RAMS:

- [Split Brain - RAMS & Validation AT - Confluence.AT](#)

List of identified VIL interfaces (intra-VIL and inter-DIL) and proposed virtual networks:

- [List of VIL interfaces and qualities](#)

TAS Plf support tickets to clarify certain NMR / split-brain behavior and configuration:

- Clarification regarding NMR on 2.6.x with network failure: <http://plf-support.at.gts/otrs/customer.pl?Action=CustomerTicketZoom;TicketNumber=2026021610000042>
- Split-brain in NMR: additional supervision channel: <http://plf-support.at.gts/otrs/customer.pl?Action=CustomerTicketZoom;TicketNumber=2026021610000051>

Abbreviation	Phrase
MCI	Master Confirm Interface
MES	Master Election and Sort
MEP	Master Election Process
MTP	Master Token Protocol
SB	Split brain
VIL	Virtual Interlocking

Motivation

Customer requirements of the DSTW requests a seamless switch-over in case of a outage if one Geo Redundancy side. In case of a total power outage of one Georedundant site or a double fault of the network between the two sides - standard switch over mechanisms of NMR switch over will not work and additional measures are required to fulfill the customer requirement.

On request of the DSTW, TAS Plf has provided additional measures to reduce the probability of a failed switch over but that increases to probability of a split-brain situation. With that EPIC the balanced solution (maybe with additional measures) should be defined and realized.

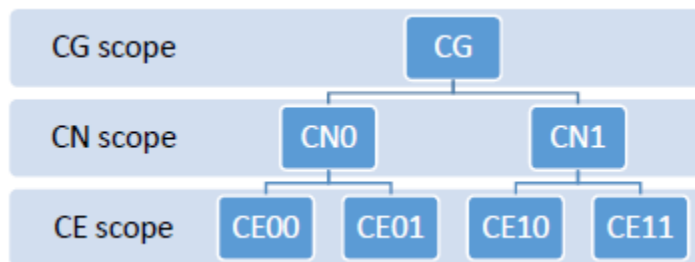
At least the following failure scenarios must be considered

- total outage of one geo-redundant site (e.g. Power-off of one side)
- double fault of the network between the two sites (split brain situation is possible) - (what's the probability of that situation?)
- double fault of network in one geo-redundant site - (split brain situation is possible) - analyze all possible failures
- outage of (e.g. Power-of) of a total availability zone - with the active Master in that zone - seamless switch-over should be possible
-

Define the proper and the best balanced solution for the VIL/DSTW in that kind of failures - split brain situations can lead to a hazard and that has to be avoided.

NMR design in a nutshell

Destination tree (scope) for NMR messages:



- Single CE - unicast
- CN - both CEs receive
- CG - all CEs (broadcast)

Leader CE roles:

- The leader CE determines the order of message processing in the CN.
- If CN is the master of CG then it determines the order of delivering application message for the slave CNs.
- Leader CE of the Master CN sends periodically tick messages to the leader CEs of the other CNs - but this message is voted and is in fact sent by the follower (see below)

Follower CE roles:

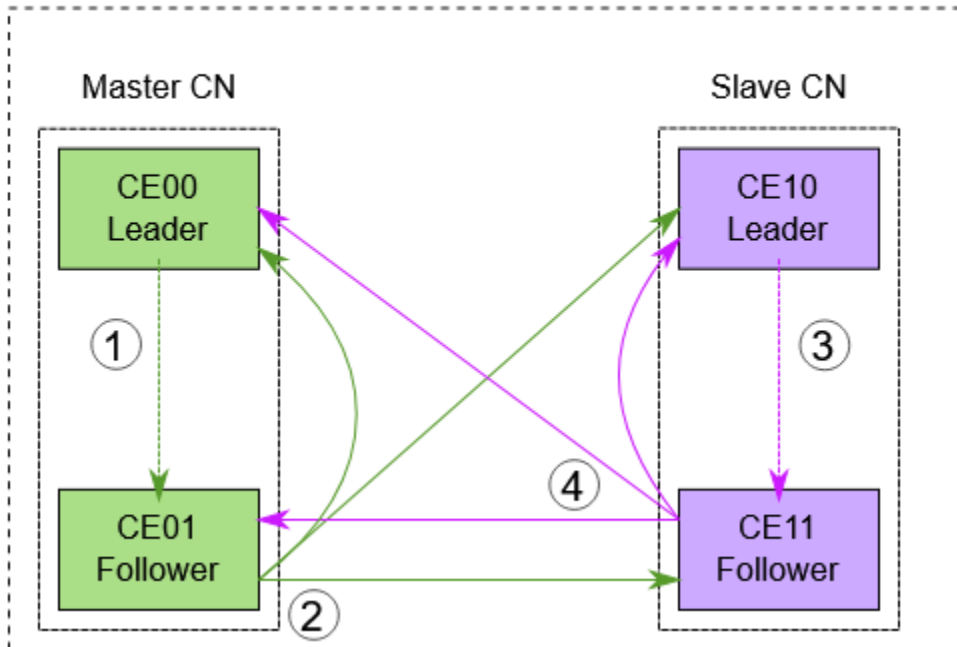
- Following the computations of the leader CE
- The progress of the leader CE is supervised through timeouts on the follower CEs in the CN



Designated sender CE:

- The follower CE with the highest CE ID
- Sends the messages after successful voting (received from the leader) to all CEs of the CG

CG



Master Election and Sort (MES):

Relevant messages from MES that handles master election and loss of the master (all messages are voted):

1. **Tick message:** Periodically sent by the master CN to the CG. It contains time progress and CG state information.
2. **Tick acknowledge message:** The tick acknowledge message is sent by all CNs to the master CN upon reception of a tick message. It informs the master CN of the current state of the CN sending this message.
3. **Master query message:** The master query message is sent by all CNs to the CG in start-up or master lost states. It requests state information of the other CNs, in particular to determine whether a master is already present.
4. **Master reply message:** The master reply message is sent by each CN to the CN from which the master query message originated. It contains the last CG state information a CN has about the CG, including whether a master is present.

Relevant timeouts (that triggers new master election process or safe state/panic):

1. **Master query timeout:** The master query timeout determines how long the CN waits for replies to its master query messages by the other nodes.
2. **Tick timeout:** The tick timeout defines the period with which the master CN sends out the tick messages.
3. **Master lost timeout:** The master lost timeout defines how long the slave CNs wait between ticks until they assume that the master has been lost and start master election.
4. **Initial master election timeout:** The initial master election timeout defines how long the process of electing a master can take from the startup of a CN, until the CN resets.
5. **Master switch timeout:** The master switch timeout defines how long the process of electing a new master after a master has been lost can take, until the CN resets.

How slave CNs assume the master is lost:

1. **No "MES Tick Message"** received from the master during the given time period (master lost timeout). The message is sent from the leader CE to the entire CG.
2. **During CN startup** if no master is elected (or seen to be elected) within given timeout
3. **Master switch timeout** when the process of electing a new master takes too much time

TAS Plf measures

- [SDD - TAS Platform Core SW - 2.x -NMR](#)

1. High availability at the communication network level

Robust infrastructure that guarantees no single point of failure. This has to be ensured at all communication layers e.g., multiple communication channels between CNs and CEs.

2. Master Confirm Interface (MCI)

API in NMR that supports different custom detection mechanisms. Before CN becomes a master, the application is requested to determine whether this is safe to do so or not.

3. Master Token Protocol (MTP)

The master token protocol solves part of the split-brain problem by introducing a master shutdown message as token to identify the cases where a new master can be safely elected. It is based on the assumption that at least one CE from the failed master CN can announce it to the remaining CN slaves.

If the CN that is about to become master did not receive these shutdown messages, regardless of whether they were not sent or lost during transmission, the master token protocol does not permit the CN to become master.

The protocol uses 2 variables:

- `mt_cn_state`, storing the last CN states when a master election is started
- `lost_master_id`, storing the id of the master CN that sent a master shutdown message



CN will become the master (after detecting master CN is lost) only if it has received shutdown message.

4. Supervision channel

[Additional Supervision Channel in Programmers Guide](#)

Requires additional independent channel through which master CN distributes "tick information" besides the NMR sync network. Slave CNs store this tick information in `adsv_cn_state`. If master's tick from this channel does not match tick information from NMR synchronization network it indicates that not all CNs were connected.

Network is not partitioned if `adsv_cn_state` (from supervision channel) is equal to `mt_cn_state` (from MTP via NMR synchronization network when master election process starts). If network is partitioned then the master has different view than the rest of the network. If views are different, then decision is forwarded to the next check (if configured) or to the MCI.



CN allowed to become a NMR master only if the connections between the additional supervision channel and the synchronization network are lost simultaneously (it assumes total loss of the "location" or a triple failure).

Communication in the supervision channel is UDP based with multi-cast addressing (unicast is possible too).

5. Observer for start-up

[Observer at Startup in Programmers Guide](#)

Is used to prevent split-brain situation during the NMR startup to detect that there is no Master CN already running. There is a component Observer Module deployed on each CE. The Observer component itself has to be deployed where it is reachable from all CNs (and Observer Modules) and must not be directly allocated with one of the CNs in particular.

The Observer application can be also [launched in the container](#).

Split-brain prevention

The objective is to:

1. To find a reliable mechanism that can detect the precondition for the split brain. That means to distinguish between two situations:
 - a. CN master failure - it is safe that remaining slaves to elect a new master CN
 - b. Partitioned network (isolated islands) - it is unknown whether the master is still fully operational (sends commands and receives messages from the OCs and/or adjacent systems) or not.
2. In case of the precondition for the split brain - to define a procedure how to prevent full-blown development of the SB and what should be the safe reaction.
3. Recovery from the precondition and from the full SB situation.

Delegation pattern in TAS Plf to prevent split-brain through multiple-checks: if one check cannot reliably determine new master than the decision is further delegated to the next check. Sequence of checks to reliably determine new master

1. Master token protocol
2. Supervision channel (optional)
3. Observer for start-up (optional)
4. MCI (optional)

Initial model of the redundant system for split-brain analysis

The purpose is to perform initial "FMEA" analysis and identify scenarios with undesired effects on safety or availability of the "default" NMR setup and without additional measures. Afterwards, various measures (incl. those from TAS Plf) will be then applied and check whether can resolved concerns of identified scenarios. ECMUX is not considered in the initial analysis and will be decided later.

The projected network topology is similar as the one in [OMC 3.x Core Network Design](#) with two Top-of-rack switches.

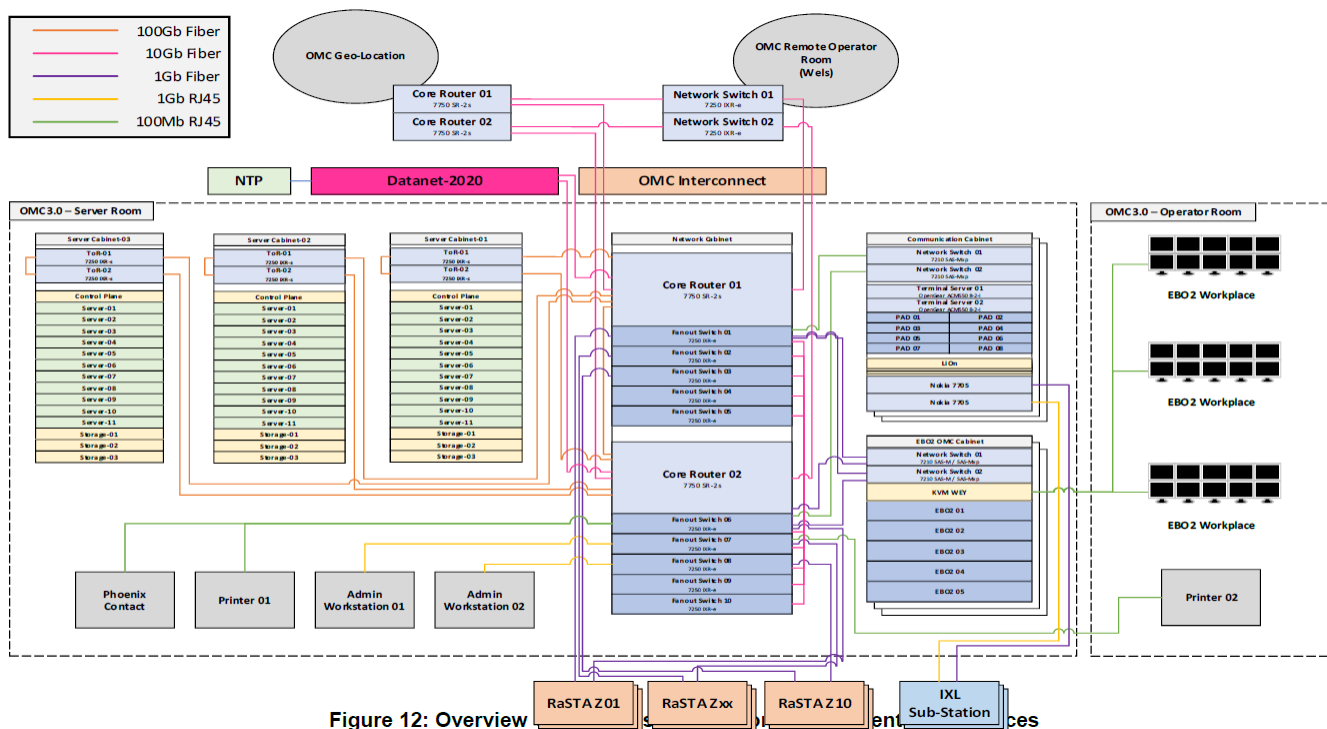
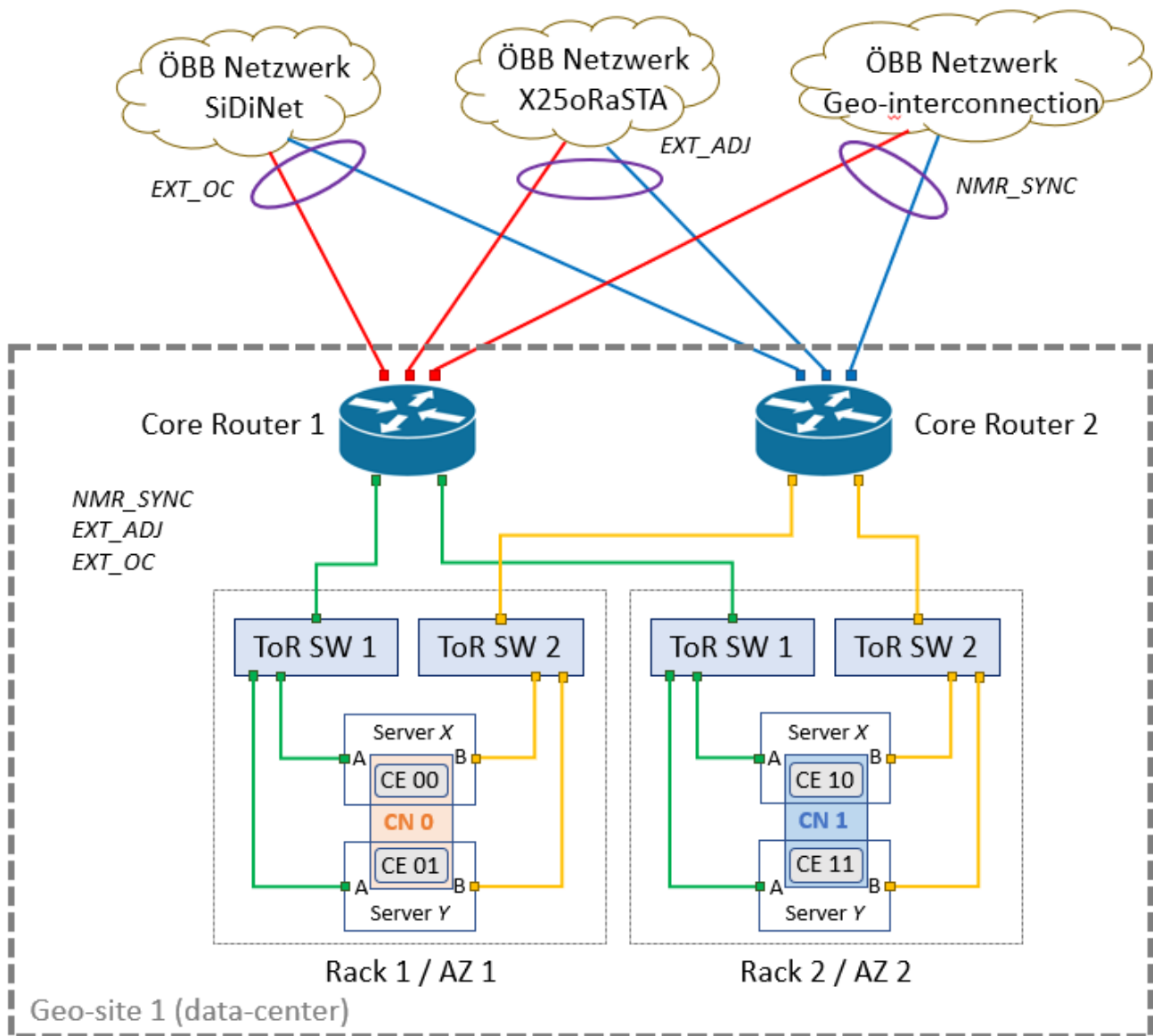


Figure 12: Overview

The model of the redundant system for initial analysis:



Assumptions:

- ECMUX is not considered in this initial analysis
- One CN is inside one availability zone (rack)
- CEs of the same CN are deployed on different server blades
- There is at most one switch between two CEs of CN (SDD NMR and network requirements)
- "IPS" Network for other interfaces (SECLOG, SYSLOG, SECMT, DLG ...) not relevant for this initial analysis, thus not shown and considered
- Bonded interface(s) for IPS control plane not relevant for this initial analysis, thus not shown and considered
- Single point of failure between communication points is excluded due to the network topology design
- Double failures are considered

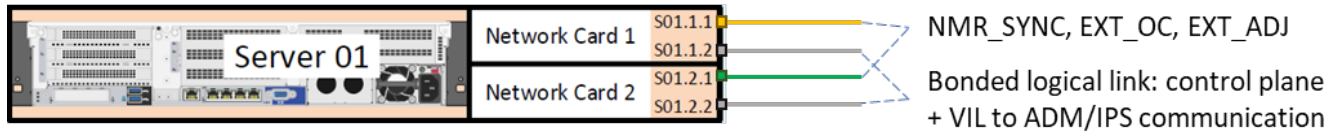
The initial analysis: [Split-brain-analysis.pptx](#)

Proposal for the split-brain avoidance

Analysis of the proposal: [Split-brain-proposals.pptx](#)

Port allocation on server blades

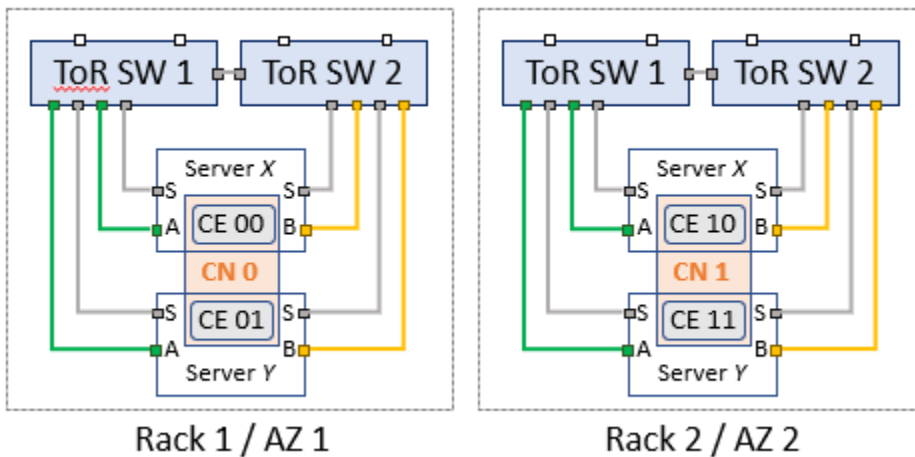
- Each server blade has 2 physical NIC with 2 ports with the following mapping:



- 2 NIC ports (yellow and green) for independent physical links used for redundant networks where applications require two independent physical channels (NMR synchronization, communication with OCs, communication with adjacent systems etc.)
- Remaining 2 NIC ports form so called **bonded logical link** (grey, S) – for control plane, and service (IPS) networks (VIL to ADM/IPS communication). The list of interfaces mapped to this network can be found in the [List of VIL interfaces and qualities](#)

Cabinet topology

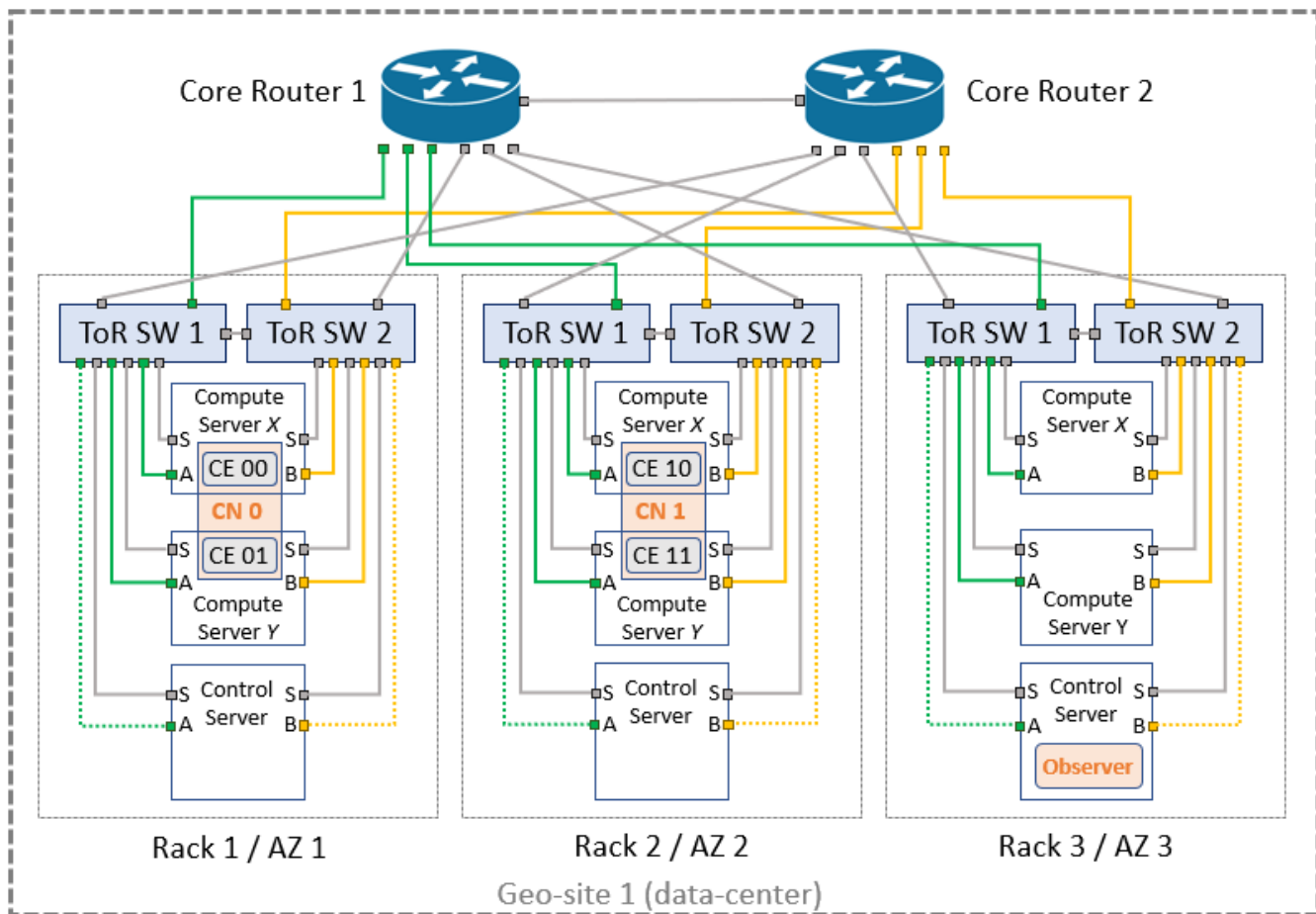
2x top-of-rack switches (preferred):



Architecture with 4xToR switches is also possible and further increases availability of the system. Rack with 2xToR switches is currently considered as sufficient (after preliminary RAM calculations).

Geo-site topology

After evaluation of various topologies the "cross" has been found as the optimal:

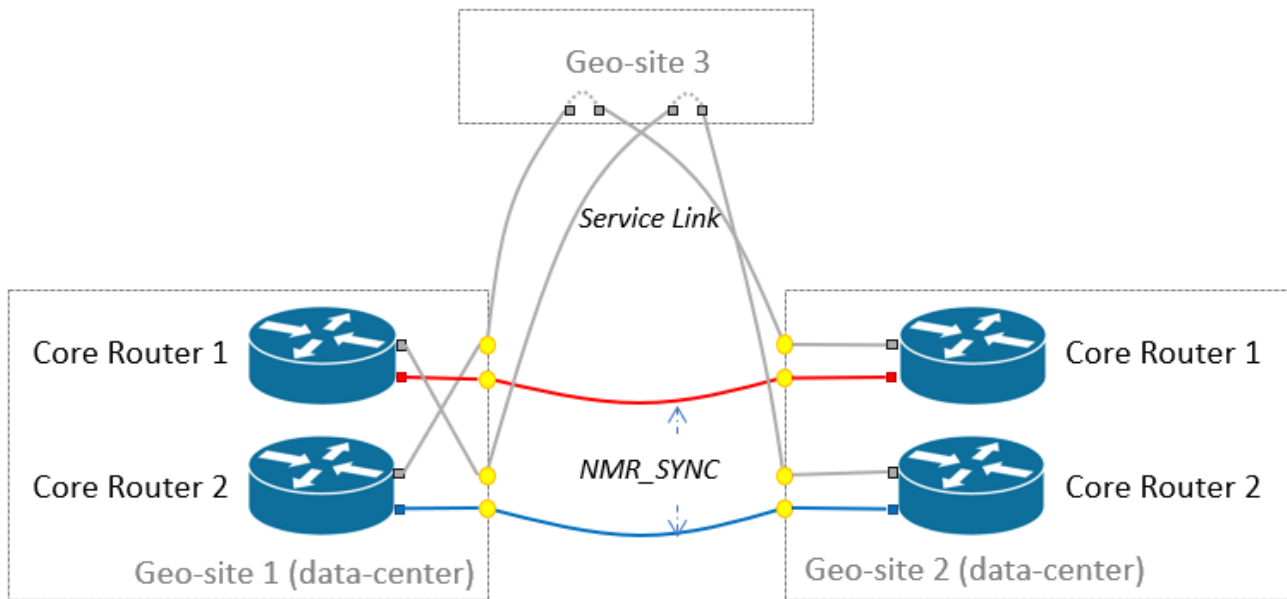


- TAS Ptf Supervision channel (Sv) is deployed on bonded logical link
- Observer application is deployed in a container in availability zone (AZ) where are no CNs of the given CG running
 - Increased availability in case of CG start, quick CN recovery
 - For basic redundancy (one geo-site) there is one AZ where the Observer for the given CG can be deployed.
 - In case of the full geo-redundancy we have 2 AZs where the Observer can be deployed. Only one site has active Observer application. If the site with active Observer fails, another container with Observer application is re-deployed in healthy (running) site. This can be automatically handled by Kubernetes and its self-healing mechanism. Similar approach will be used for ADM services (that have to be after one site failure re-deployed too). ~~6 AZ (3x2), CG is deployed in 4. That leaves 1 AZ at each geo-site for potential Observer deployment~~
 - If the geo-site where is currently deployed Observer app is due to fault(s) unavailable, the container with the Observer application is deployed in the other geo-site (similarly as with all ADM services)
- We need only **minimal MCI module** that will always return "not okay" for the master conform request. ~~No need for the custom application at MCI~~

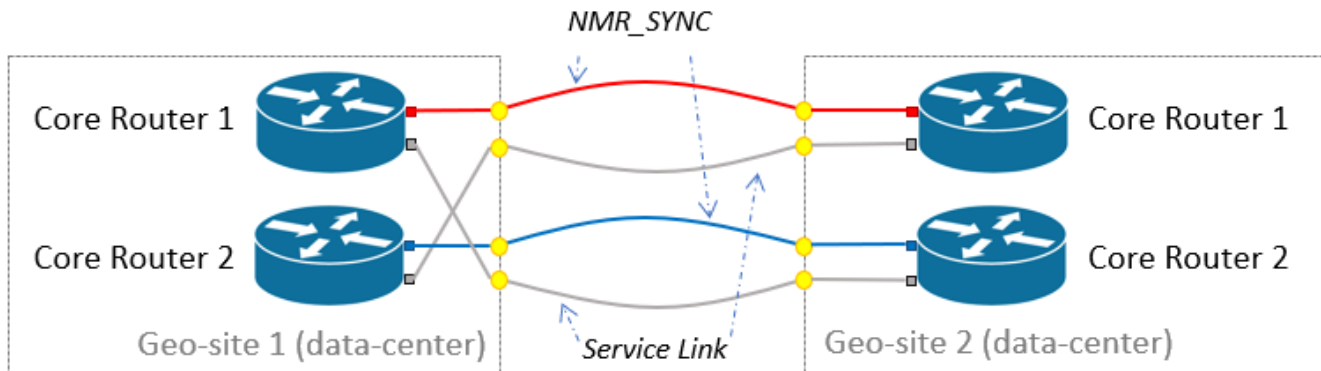
Interconnection of geo-sites

There will be four independent physical links on each geo-site provided from ÖBB. It is the same infrastructure as in the OMC project. However, OMC aggregates those four links into two independent logical rings between three geo-sites (not our case). In DSTW will not use ring topology. ~~two independent physical rings from ÖBB~~. It has not been decided if provided four links will include 3rd geo-site (similar as in OMC project) or not. Thus there are two options that are functionally identical:

Option 1:



Option 2:

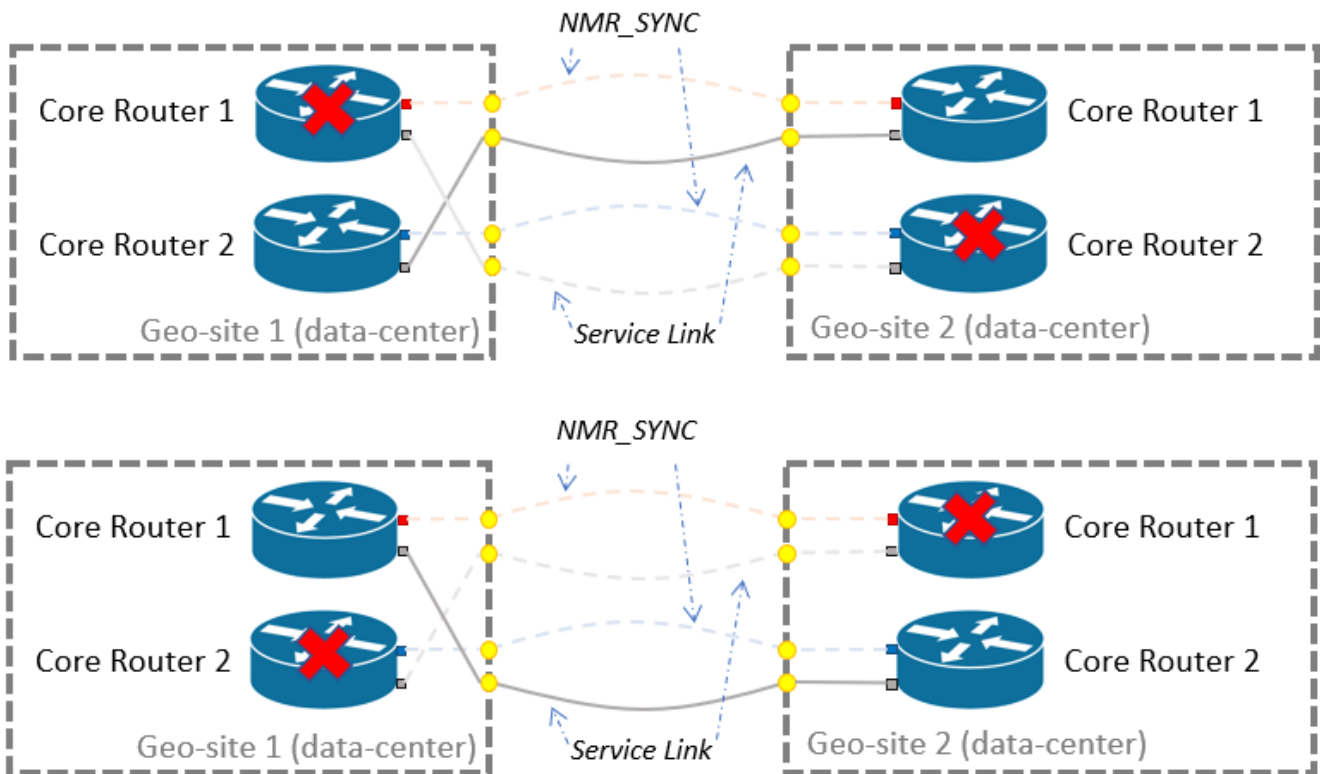


Geo-sites are interconnect with four physically independent links provided by ÖBB:

- Service link (gray) is logical bonded connection that has the same meaning as inside the geo-site: control plane, supervision channel, deployed observer, communication between VIL and IPS/ADM services.
- Physically redundant links (red and blue) for NMR synchronization network

Both options are functionality equivalent. The service link in the *Option 1* might have higher delay due to an extra hop but this strongly depends on the provider's (ÖBB) quality of service.

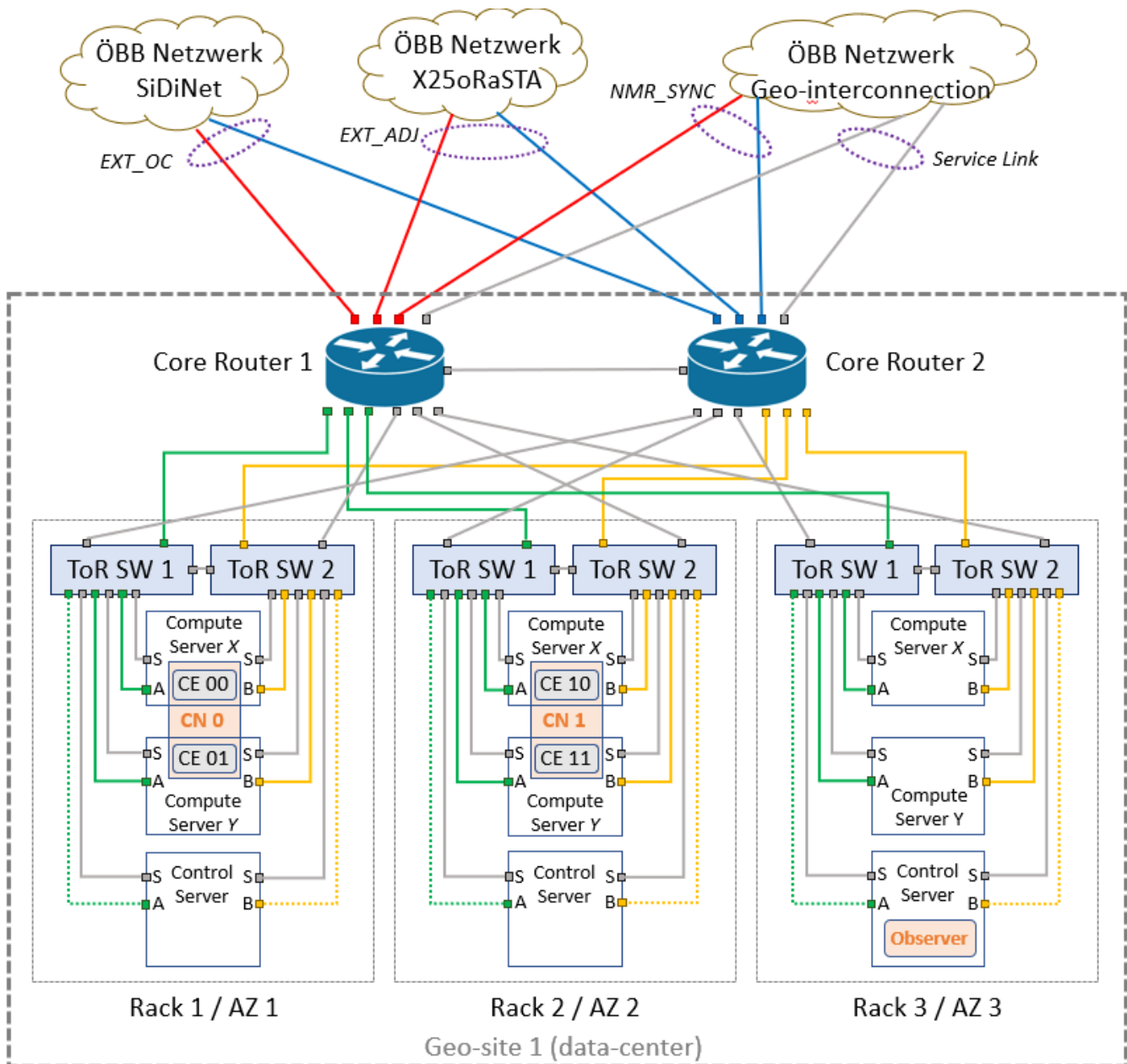
Yellow points are interfaces between customer's network and our "rack". One geo-site will have a "crossed" connection to eliminate split-brain in the following double fault situations (service link is still available, thus supervision channel will prevent election of the 2nd master):



If four independent physical links provided by ÖBB are not sufficiently available then it is possible to establish additional path (link) between geo-site via ÖBB RaSTA Netzwerk as an alternative.

The proposed architecture

The following figure shows the overall infrastructure architecture of the proposed split-brain avoidance.

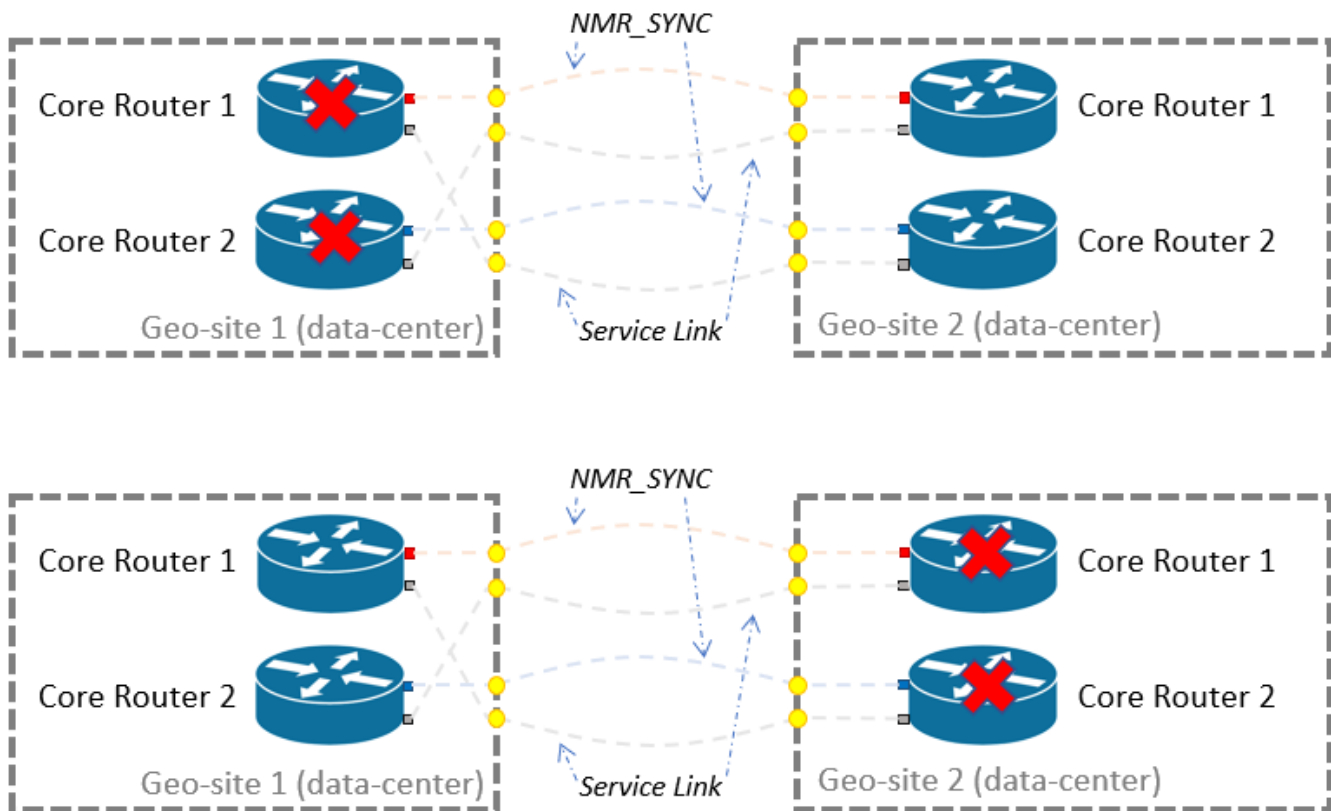


Summary of the proposed solution

The proposal handles the following failures:

- Any kind of a single fault
- Power outages within availability zone (rack) as well as the entire geo-site
- Any double (or even triple) link faults between geo-sites without a safety hazard
- Certain combination of double faults might affect only availability
- Double fault of core routers in any combination is neither a safety hazard nor availability issue. However, **hazard might occur only during the reintegration** (see below for clarification) of a faulty core router.

The following double faults can be a safety hazard due to the split brain during reintegration (recovery of one of the faulty routers):



If dual faults on core routers happen, one site is fully isolated from any external communication (not only with the respective geo-site). Thus, if a new master is elected in the isolated site it cannot communicate with OCs and adjacent systems. ~~thus not advancing computation.~~ Furthermore, if both core routers are down, communication between racks (AZs) within the affected geo-site is down too. Thus, several services and interfaces might become unavailable to CEs. One of the consequences is that communication with CDS is lost too. After CDS communication link timeouts (*TBD how long, minutes?*) CEs will automatically reset. Since there is no connection to the CDS server then booting CEs will not be prevented to get operational (*TBD how it happens*).



The safety hazard happens if one of the faulty routers becomes suddenly available - the NMR will ensure that only one master exists (with the lowest ID) while the other master will reset. Might happen that the master with the lowest ID is the one that was isolated and has an old state. This hazard can be prevented with an operational procedure. ~~if one or two routers fails, the affected site has to be disconnected, all CNS shutdown. After the site infrastructure is back and online, CN can be again started.~~

System failure rates for the above independent double fault of core routers in one geo-site (assuming **10.000 FiT** per core router) considering different time to repair after one fault occurs (see Update 18.3.2026 in [Split Brain - RAMS & Validation AT - Confluence.AT](#)):

MTTR [h]	_sys (FiT)	MTBF [y]	Availability	min/year down
1	0.20	570 776	99.999 999 98	0.0001
4	0.80	142 694	99.999 999 68	0.0017
8	1.60	71 347	99.999 998 72	0.0067
12	2.40	47 565	99.999 997 12	0.0151
24	4.80	23 782	99.999 988 48	0.0605
72	14.40	7 927	99.999 896 32	0.5449

Green color is within SIL4 (hazard rate 10E-8 to 10E-9).

Failure rate for Common-Cause Failures (CCF) with conservative = **0.02** and **10,000FiT per router**, the CCF path contributes **200FiT**,



Common-Cause Failure of both core routers does not create an immediate hazard because the affected site is isolated. The hazard might happen during "re-integration" of a single core router. Nevertheless, this shall be addressed in the operational procedures. To be determined by the customer whether this CCF rate is acceptable or assumptions (factor) shall be different. To further eliminate CCF, it is possible to introduce second pair for routers dedicated for the supervision channel and communication with the Observer application.

Requirements for VIL

1. NMR config: MTP shall be enabled
2. NMR config: Supervision channel shall be enabled and bind to the network where all CEs of the CG are reachable. Transport protocol is UDP and addressing via multi-cast
3. NMR Observer module for start-up shall be loaded and configured to connect to the Observer application (hosted by IPS). Hostname (address) and UDP port to be agreed out-of-band with IPS.
4. Master Confirm Interface shall have attached "minimal" script with "Not_OK" reply on the master proposal and it shall log the info in syslog, e.g. [Minimal Example of Master Confirm Module](#)

Proposed requirements for IPS core network topology

Physical network topology is assumed to be as described in "Proposal for the split-brain avoidance"

1. One bonded link used for supervision channel, communication with Observer and other networks for VIL to ADM/IPS communication
2. Two independent links used for redundant networks: NMR sync, external communication with OCs, external communication with adjacent systems
3. Geo-interconnection shall have a "crossed" bonded link on one site to improve availability in case of certain combinations of double faults on core routers

Proposed requirements for IPS to deploy Observer Application

1. There shall be exactly one Observer application instance per each VIL. Multiple instances might be a hazard (to be double checked).
2. Each VIL (VMs forming CG) shall have access (connection) to the Observer application via so called "ServiceLink". CNs in both geo-locations shall communicate with the same Observer app.
3. Communication protocol between VIL and the Observer is proprietary based on UDP. In order to reduce configuration overhead, all VILs shall have the same configured address/hostname of the Observer application.

Proposed ICD content for IF-IPS-OBS interface

1. Communication via ServiceLink (bonded logical link)
2. [Observer application is provided as a container with the following parameters:](#)
 - a. Observer Name - provided by VIL
 - b. Observer ID - provided by VIL, the same ID has NMR config
 - c. UDP port binding - agreed with IPS, must be the same as in NMR config
 - d. Timeout - provided by VIL
3. Observer replies to UDP datagrams (source IP address)
4. Each CE on the "ServiceLink" shall be assigned distinct address (minimum within VIL scope) via DHCP mechanism (generic point, independent of the observer)

Communication performance

1. **Regular traffic:** Each 3rd time tick (300ms) all CNs (i.e. 8 CEs) send 1 UDP datagram (OB_STATUS) to the Observer application with the payload size 26B (OB_STATUS message).
2. **Sporadic traffic:** When VIL starts or a new master is being elected. This round can happen many times within the time period defined as "Initial Master Election Timeout" or "Master Switch Timeout":
 - a. All alive CEs send request (OB_STATUS_REQUEST) to the Observer application of size 20B. The worst case: 8x 20 bytes.
 - b. Observer replies to each CE with the message size (UDP payload) of 86 bytes (OB_STATUS_REPLY), the worst case 8x 86 bytes
3. No other traffic between VIL and Observer application is expected

Application performance

The application has a very low processing intensity that is event-driven:

1. Reactive (event-driven) processing: Application is active only when processing messages from CEs:
 - a. OB_STATUS : stores the most received message per each CE into an array
 - b. OB_STATUS_REQUEST: Replies to the sender with OB_STATUS_REPLY that is the array with stored OB_STATUS messages (see above).
2. Internal continuous processing (between messages): None. There is only timer which is reset when OB_STATUS is received. If timer expires, application updates only a state variable. No other processing happens.

Other qualities

Very high availability: Observer plays critical role during starting the station. In principle, it is not possible to seamlessly start a CG without Observer application being accessible from all starting CEs.

Health (availability) of the container with the Observer application shall be monitored and the container shall be re-started (clean state) if not responsive, fault, or dead. The container restart **shall follow deployment constraints**.

Deployment constraints

1. Container for a given VIL (station) shall be deployed in AZ that does not host CN(s) of the same VIL.
2. At any moment of time, there **shall be at most one active** (visible, operating) Observer application per station. Having no Observer application during a short period of time (minutes, maybe hour) is also acceptable. Two active instances at the same time **shall not** happen (e.g. no load-balancing).
3. The Observer application for the station(VIL) shall be accessible under the same hostname/IP address and UDP port from all CEs of the given station.
4. Location of the Observer app deployment (AZ/geo-site) **shall be fully transparent** to the entire VIL. No change of VIL (CEs) configuration if the container is re-deployed or relocated to another AZ or geo-location (regardless the reason).

Foreseen failure scenarios that must be considered and expected outcome for Observer Application

1. Double fault of core routers in one geo-site: The entire affected site is not accessible, no communication between AZ in the affected location. CNs from the other, unaffected geo-site shall eventually have access (communicate) with the Observer application that is re-deployed (clean state). It shall be prevented that if the affected geo-site will become again connected (available) there will be maximum one Observer application instance (per station) accessible from both geo-locations.
2. Double fault on ServiceLink between geo-location: VIL continues operation unaffected but the Observer is unavailable for one site. It shall be prevented that there will be active two Observer applications in both geo-locations. For instance, if there is an automatic re-deployment of the Observer, it should be prevented that in this situation the container controller will re-deploy the second container.

Follow up discussions with stake holders

IPS

- Requirements:
 - [2026-05-12 IPS Split-Brain requirements - Product Engineering AT - Confluence.AT](#)
- Technical feasibility of the proposed core network infrastructure:
 - [2026-05-08 Meeting notes - VIL-IPS Infrastructure changes for split-brain avoidance](#)
- Observer deployment:
 - [2026-05-06 Meeting notes - VIL-IPS Observer deployment - Introduction - Product Engineering AT - Confluence.AT](#)
- HW qualification:
 - tbd

Security

- tbd

RAMS

- [Split Brain](#)

